UChicago REU Notes Week 4

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Remark 0.1. Assume everything is compactly generated, so we have a natural homeomorphism $\mathcal{U}(X \times Y, Z) \cong \mathcal{U}(X, \mathcal{T}(Y, Z))$, and $\mathcal{T}(X \wedge Y, Z) \cong \mathcal{T}(X, \mathcal{T}(Y, Z))$. Here, $\mathcal{U}$ are compactly generated spaces, $\mathcal{T}$ are based spaces.

For basic reviews of G-spaces, I'll type up the notes later.

1 Basics of Equivariant Homotopy

2 Motivation of Categorical Aspect

Goal: Understand to construct genuine G-spectra. A naive G-spectra is a spectra with a G-action (suposedly the structure maps are all G-equivariant + homotopy equivalence, or homeomorphism).Seems like we require extra structures on the spectra. One of the distinction seems to be: naive G-spectra represents $\mathbb{Z}$-graded cohomology theories, genuine G-spectra are used to represents cohomology theories for real representations (so it's also encoded using representations). I particular, understanding what is the equivariant infinite loop space machine (construct from G-space / category, forming a G-spectra).

Definition 2.1. Given a category $\mathcal{V}$, $X \in \mathcal{V}$ is a G-object if it has a group homomorphism $G \to \mathrm{Aut}(X)$.

$\mathcal{V}_G$ denotes the category of G-objects, with morphisms being in $\mathcal{V}$.Each $\mathcal{V}(X, Y)$ has a natural G-action by conjugation, where $g \in G$ acts as $f \mapsto g \circ f \circ g^{-1}$.

$G\mathcal{V} \subseteq \mathcal{V}_G$ denotes the subcategory where homsets are limited to G-maps (so $G\mathcal{V}(X, Y) := \mathcal{V}(X, Y)^G$, fixed points of the homsets). Another notation is $G\mathcal{V} := (\mathcal{V}_G)^G$.

Definition 2.2. Given $X \in G\mathcal{T}$ (based G-space), assume inclusion of basepoint $* \to X$ is nondegenerated in the G-case, meaning it's a G-cofibration (satisfying G-homotopy extension property for G-equivariant maps, abbreviated as G-maps). In particular, all the homotopies in the extension property is required to be a G-map at all time t (another way of phrasing, interval I has trivial action, so action at slice $X \times t$ is equivalent to action on $X \times I$).

By chasing through the homotoopy extension diagram (by restricting all homotopies to be G-homotopy), then inclusion of basepoint $* \to X^H$ becomes a cofibration with respect to G-maps (Note: Not necessarily a G-cofibration, because in general X^H is not closed under G-action). However, since all G-space is naturally an H-space (so is homotopy), then we can understand this as "H-cofibration", as H acts trivially on X^H :0

Definition 2.3. A G-map $f : X \to Y$ is a weak equivalence of G-spaces (or G-equivalence), if the fixed point maps $f^H : X^H \to Y^H$ are weak homotopy equivalences for all subgroups $H \leq G$.

Similar G-whiteheads theorem states if X, Y are G-CW complexes, then f is also a G-homotopy equivalence.

Definition 2.4. A small category $\mathcal{C}$ is a *topological category*, if the set of objects Ob and set of all morphisms Mor are topological spaces, such that the following four maps are continuous:

- Identity map $i : \mathrm{Ob} \to \mathrm{Mor}$ by $i(x) := \mathrm{id}_x$ is a continuous map.
- Source map $s : \mathrm{Mor} \to \mathrm{Ob}$ by $s(f : X \to Y) := X$ is continuous.
- Target map $t : \mathrm{Mor} \to \mathrm{Ob}$ by $t(f : X \to Y) := Y$ is continuous.
- Composition $\circ : \mathrm{Mor} \times_{t,s} \mathrm{Mor} \to \mathrm{Mor}$ is continuous.

Which, the last one is collecting all pair (X, Y) with morphism $f : X \to Y$ between them.

In particular, s, t are retracts of i, showing i is an embedding. (check https://mathoverflow.net/questions/85717/segals-original-definition-of-a-topological-category)

There is another notion enriched over $\mathcal{T}$.

Notation 2.5. Let Cat denotes the category of small topological categories, which Cat_G denotes the topological G-categories (meaning G acts on both $\mathrm{Ob}, \mathrm{Mor}$ such that $i, s, t, \circ$ are all G-maps), an equivalent formulation is a group homomorphism $G \to \mathrm{Aut}_{\mathsf{Cat}}(\mathcal{C}, \mathcal{C})$, by checking the definition.

Part of the reason we endow this definition, is so that for subgroup $H \leq G$, the *H-fixed category* $\mathcal{C}^H$ will make sense, namely we need the action on $\mathrm{Ob}, \mathrm{Mor}$ to be compatible.